BLOCKCHAIN-BASED STUDENT ATTENDANCE MANAGEMENT SYSTEM

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Use Case Report

submitted by

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(An NBA & NAAC accredited and ISO 9001:2015 certified institute)

Kanuru, Vijayawada-520 007

2026-27

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CERTIFICATE

This is to certify that **Chagantipati Chaitra (23501A0522)** of **IV B.Tech (4-1), Department of Computer Science and Engineering**, has successfully prepared and submitted the Use Case Report titled **"Blockchain-Based Student Attendance Management System"** in fulfillment of **Assignment–2** requirements for the academic year **2026–2027**, as part of the Bachelor of Technology curriculum.

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1. INTRODUCTION

1.1. Overview of Blockchain Technology

Blockchain technology is a decentralized and distributed ledger system that records transactions across multiple nodes, ensuring transparency, security, and immutability. Unlike traditional databases, blockchain operates on a peer-to-peer network, eliminating the need for central authorities. Each block in the chain contains a cryptographic hash of the previous block, making data tampering virtually impossible. Originally designed for cryptocurrency transactions, blockchain has now expanded into various domains, including finance, supply chain, healthcare, and digital warranty management.

1.2. Relevance of Blockchain in Student Attendance Management Systems

In educational institutions, attendance is an important activity used to monitor student participation in classes. Traditional attendance management systems may use paper registers or centralized databases. These methods can involve manual errors, difficulties in maintaining records, and challenges in reviewing previous attendance entries.

A Blockchain-Based Student Attendance Management System uses a smart contract to record attendance transactions. In this project, the administrator can register students and authorize faculty members. Authorized faculty members can mark students as present or absent for a selected subject.

The system is developed using React for the frontend, Solidity for the smart contract, Hardhat for the local blockchain environment, MetaMask for wallet connection, and Ethers.js for communication between the frontend and the smart contract.

The application retrieves student information and attendance records from the smart contract and displays them through the dashboard. It also provides attendance history and percentage reports based on the recorded attendance entries.

The project demonstrates how blockchain technology and smart contracts can be used to maintain traceable attendance records and enforce permissions for specific operations. The current implementation runs on a local Hardhat blockchain network and serves as an academic prototype for blockchain-based attendance management.

The system also helps reduce the effort required to maintain and review attendance records. By combining a web interface with blockchain-based record storage, the project provides a simple and traceable approach to attendance management in educational institutions.

2. BACKGROUND

Traditional attendance management systems face several challenges in maintaining student attendance records. Many institutions use paper registers or centralized databases to record attendance. These methods may require manual effort and can make it difficult to review and verify previous attendance records.

2.1. Manual Attendance Management

In traditional systems, faculty members record student attendance manually. This process takes time, especially when handling multiple students and subjects. Manual entry may also lead to mistakes while recording or maintaining attendance information.

2.2. Difficulty in Maintaining Records

Maintaining attendance records over a long period can be difficult. Paper records may be misplaced or damaged, while stored records may require additional effort to retrieve and verify. A structured digital system helps organize student details and attendance information.

2.3. Access Control

Attendance records should be managed by authorized users. If permissions are not properly controlled, unauthorized users may attempt to perform attendance-related operations. Therefore, controlling access for administrators and faculty members is an important requirement**.**

2.4. Lack of Traceability

Reviewing previous attendance entries can be challenging when records are not maintained in a structured manner. It is useful to have attendance information that includes the student ID, subject, status, timestamp, and the wallet address that submitted the transaction.

2.5. Project Approach

The Blockchain-Based Student Attendance Management System addresses these requirements using a Solidity smart contract. The administrator can register students and authorize faculty wallet addresses. Authorized faculty members can mark students as present or absent for a selected subject.

The application retrieves attendance records from the smart contract and displays attendance history and percentage reports through a React dashboard. The project uses Hardhat for the local blockchain environment, MetaMask for wallet connection, and Ethers.js to connect the frontend with the smart contract.

The current implementation is an academic prototype that runs on a local Hardhat blockchain network

2.6. Data Integrity and Reliability

Maintaining accurate attendance records is important for educational institutions. In traditional systems, attendance information may be changed accidentally or modified without proper authorization. Blockchain technology helps improve data integrity by recording transactions on a distributed ledger. Once a transaction is confirmed, changing the recorded information becomes difficult. In this project, attendance details are stored through a Solidity smart contract, helping maintain a traceable record of attendance transactions.

2.7. Role-Based Access Control

Access control is necessary to ensure that only authorized users can perform specific operations. In the proposed system, the administrator is responsible for registering students and authorizing faculty wallet addresses. Only authorized faculty members can mark attendance for students. The smart contract checks permissions before allowing these operations. This helps prevent unauthorized users from modifying attendance records and improves the reliability of the system

2.8. Attendance Tracking and Reporting

Tracking student attendance is essential for monitoring academic participation. Traditional methods may require faculty members to calculate attendance percentages manually. The proposed system provides attendance history and percentage reports through a React-based dashboard. Attendance records can be reviewed based on student details and subjects. The system uses the recorded attendance entries to calculate attendance percentages, making it easier to review student attendance information.

2.9. Need for a Blockchain-Based Solution

A blockchain-based attendance management system provides a structured approach to recording and reviewing attendance. By combining smart contracts with a web application, the system supports permission-controlled attendance entry and traceable records. React provides the user interface, Solidity defines the attendance rules, Hardhat provides the local blockchain environment, MetaMask manages wallet interaction, and Ethers.js connects the frontend to the smart contract. The current project is an academic prototype running on a local Hardhat network. It demonstrates how blockchain technology can be applied to improve the transparency and traceability of student attendance management.

3. BLOCKCHAIN BASICS

Blockchain technology is a distributed ledger system that records transactions securely and transparently. Unlike traditional systems that depend on a central authority, blockchain uses a network to validate and store transactions. Its important features include decentralization, immutability, transparency, smart contracts, and cryptographic security. These features make blockchain useful for maintaining reliable digital records. In this project, blockchain technology is used to record student attendance and manage permissions through a smart contract.

3.1. Decentralization

Decentralization means that data and operations are not controlled entirely by a single central authority. In traditional attendance systems, records are usually maintained in a centralized database controlled by an institution. Such systems may face problems if the database becomes unavailable or if records are changed without proper authorization.

Blockchain provides a different approach by maintaining records through blockchain transactions. In this project, a local Hardhat blockchain network is used to develop and test the attendance management system. The smart contract controls student registration, faculty authorization, and attendance marking. Although the current project runs on a local network rather than a public decentralized network, it demonstrates how blockchain-based attendance records can be managed.

3.2. Immutability

Immutability is one of the important features of blockchain. It means that once a transaction is confirmed and recorded, changing or removing it is difficult. Blockchain uses cryptographic techniques and links between blocks to help protect the integrity of recorded transactions.

In a student attendance management system, maintaining reliable records is important because attendance information is used to monitor student participation. The smart contract records attendance transactions containing details such as student ID, subject, attendance status, timestamp, and faculty wallet address. These records can be reviewed through the application, helping users trace previously recorded attendance information

3.3. Transparency

Transparency means that recorded transactions can be inspected by users who have the required access. Blockchain provides a transaction history that helps users review previous activities and verify recorded information.

In this project, attendance history can be viewed through the React dashboard. The system retrieves the relevant information from the smart contract and displays attendance details for review. This helps faculty members and administrators track recorded attendance entries and understand when transactions were submitted.

3.4. Smart Contracts

Smart contracts are programs deployed on a blockchain that execute predefined rules. In this project, the smart contract is written in Solidity and manages student registration, faculty authorization, and attendance marking. It checks user permissions before allowing operations. The React frontend connects to the contract using Ethers.js, while MetaMask is used to approve transactions.

3.5. Consensus Mechanisms

Consensus mechanisms help blockchain networks agree on valid transactions. Common mechanisms include Proof of Work (PoW), which uses computational work, and Proof of Stake (PoS), which selects validators according to network rules.

This project uses Hardhat to run a local blockchain network for developing and testing the attendance smart contract. It focuses on attendance management rather than implementing a consensus mechanism.

3.6. Cryptographic Security

Blockchain uses cryptographic techniques to protect transactions and verify their authenticity. Public and private keys are used in wallet-based systems to identify accounts and authorize transactions. Hash functions help protect the integrity of blockchain data by producing fixed-length values from input data.

In this project, MetaMask manages the user's blockchain wallet and is used to approve transactions. The connected wallet address helps the smart contract identify the account submitting an operation. The contract then checks whether the account has the required administrator or faculty permission.

These mechanisms support secure transaction submission and permission control. However, overall security also depends on correct smart contract design, safe wallet handling, and proper testing.

By combining these blockchain concepts with React, Solidity, Hardhat, MetaMask, and Ethers.js, the project demonstrates how a blockchain-based system can record and review student attendance. It provides an academic prototype for improving the traceability and permission-controlled management of attendance records.

4. USE CASE OVERVIEW

The Blockchain-Based Student Attendance Management System aims to simplify attendance recording by using blockchain technology and smart contracts. The system allows the administrator to register students and authorize faculty members. Authorized faculty can mark attendance for selected subjects, and the recorded details can be viewed through a web application.

4.1. Objectives

The main objective of the Blockchain-Based Student Attendance Management System is to develop a secure and efficient method for recording student attendance using blockchain technology. Traditional attendance systems may involve manual errors and difficulties in maintaining records. This project uses smart contracts to improve the reliability and traceability of attendance information.

One objective is to digitize attendance recording through a web application. Authorized faculty members can mark students as present or absent for a selected subject, reducing the need for manual registers.

The system also aims to maintain traceable attendance records. The smart contract stores details such as student ID, subject, attendance status, timestamp, and faculty wallet address, allowing previous entries to be reviewed.

Another objective is to provide access control. The administrator can register students and authorize faculty wallet addresses, while only authorized faculty members can mark attendance.

The application also displays attendance history and percentage reports, making it easier to review student attendance. React, Solidity, Hardhat, MetaMask, and Ethers.js are used to develop and test the system on a local blockchain network.

Overall, the project demonstrates how blockchain technology can support permission-controlled attendance recording and improve the traceability of student attendance records.

Another objective is to reduce the effort required for attendance management by providing an organized dashboard for viewing student details, attendance history, and percentage reports. This helps faculty members review attendance information more conveniently.

The project also aims to provide a foundation for future improvements in educational attendance systems. The prototype can be extended with additional features and tested further to support more efficient attendance management.

4.2. Scope of the System

The project focuses on recording and reviewing student attendance through a blockchain-based application. The main components include:

1. **Administrator:** Registers students and authorizes faculty wallet addresses.
2. **Faculty Members:** Marks students as present or absent for selected subjects after authorization.
3. **Smart Contract:** Applies permission checks and records attendance transactions.
4. **Web Application:** Provides the interface to manage students and view attendance history and percentage reports.
5. **Blockchain Network:** Stores transactions on the local Hardhat development network.

The current implementation is an academic prototype designed for development and testing. It does not use a public blockchain network.

4.3. System Architecture

The system architecture consists of three main layers: the user layer, web application layer, and blockchain layer. These components work together to record and display attendance information.

4.3.1. User Layer

This layer consists of the users who interact with the attendance management system.

- **Administrator**: Registers students and authorizes faculty wallet addresses.
- **Faculty Members**: Authorized faculty members mark students as present or absent for a selected subject.
- **Wallet Connection**: MetaMask allows users to connect their wallets and approve transactions.
- **Attendance Monitoring**: Users can view available attendance history and percentage reports through the web application.

4.3.2. Web Application Layer

This layer acts as the interface between users and the blockchain.

- **User Interface**: Developed using React to display student details, attendance forms, history, and percentage reports.
- **MetaMask Integration**: Connects the user's wallet and allows transaction approval.
- **Ethers.js**: Connects the React frontend with the Solidity smart contract.
- **Attendance Management**: Sends attendance requests to the smart contract and displays the retrieved attendance information.

4.3.3. Blockchain Layer

The blockchain layer manages attendance records and user permissions using a Solidity smart contract.

- **Solidity Smart Contract**: Handles student registration, faculty authorization, and attendance marking.
- **Hardhat Network**: Provides a local blockchain environment for deploying and testing the smart contract.
- **Attendance Records**: Stores student ID, subject, attendance status, timestamp, and faculty wallet address.
- **Access Control**: Verifies user permissions and allows only authorized faculty members to mark attendance.
- **Transaction Processing**: Ethers.js connects the React frontend to the smart contract, while MetaMask is used to approve transactions.

This layer helps maintain traceable attendance records and improves the reliability of attendance management.

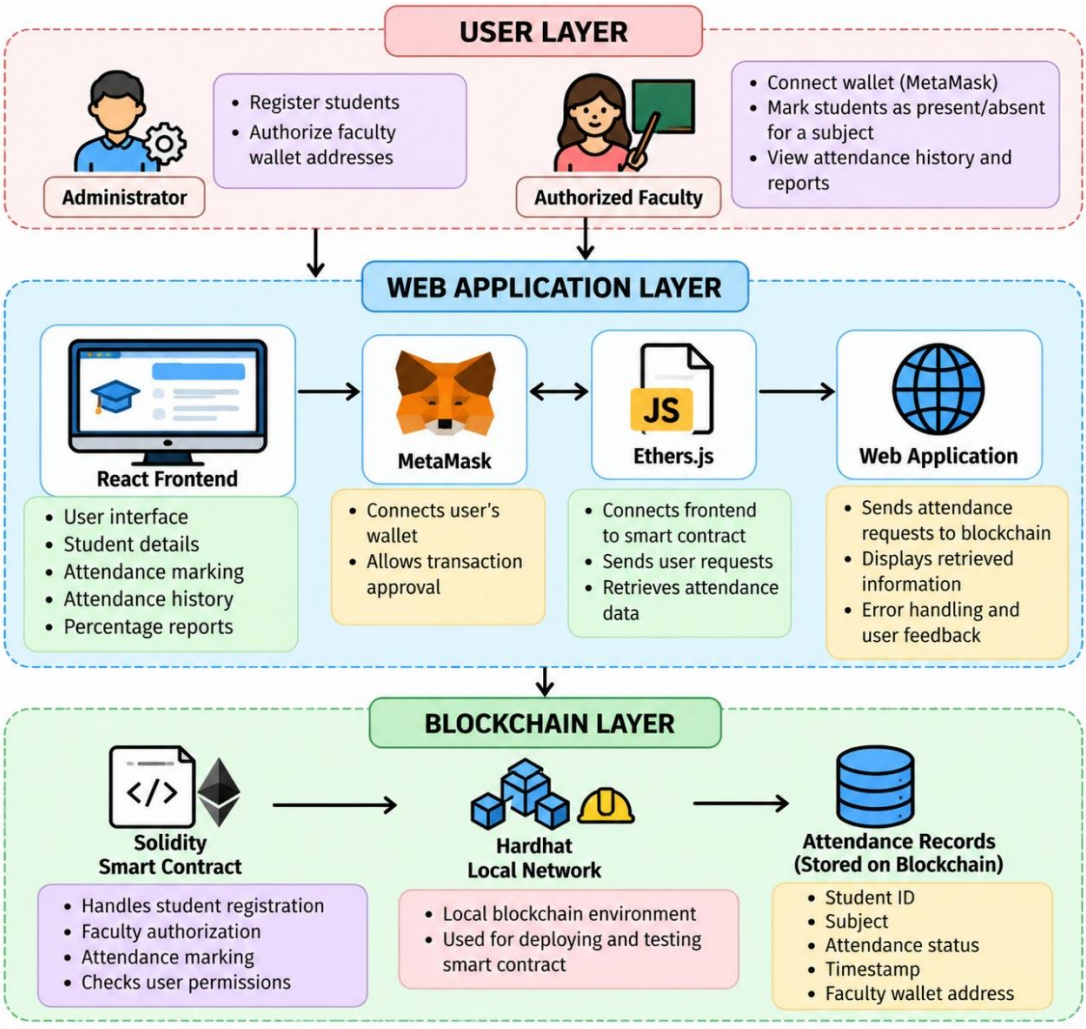


Figure 4.1: Architecture of Blockchain-Based Student Attendance Management System

5.IMPLEMENTATION

5.1. Setting Up the Blockchain Environment

The first step in implementation is setting up the development environment. React is used to build the frontend, Solidity is used to write the smart contract, and Hardhat is used to compile and deploy the contract on a local blockchain network. MetaMask manages wallet connections and transaction approvals. Ethers.js connects the frontend with the deployed smart contract.

The following tools are used in the implementation:

- **React:** Develops the user interface and attendance dashboard.
- **Solidity:** Defines the rules for registration, authorization, and attendance recording.
- **Hardhat:** Compiles and deploys the smart contract on a local blockchain.
- **MetaMask:** Connects user wallets and approves transactions.
- **Ethers.js:** Enables communication between React and the smart contract.

The basic Hardhat setup commands are:

```
mkdir attendance-blockchain
cd attendance-blockchain

npm init -y
npm install --save-dev hardhat
npx hardhat
```

After creating the Hardhat project, the Solidity contract is placed in the contracts folder. The contract can be compiled using:

```
npx hardhat compile
```

5.2. Defining Smart Contracts

5.2.1. Writing the Attendance Smart Contract

The smart contract contains the main rules of the attendance system. It manages student registration, faculty authorization, and attendance records. Each attendance record contains the student ID, subject, attendance status, timestamp, and faculty wallet address.

A structure is used to organize attendance information. Mappings help identify registered students and authorized faculty members, while an array stores attendance entries.

Sample Solidity Code:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

contract Attendance {
    address public admin;

    struct AttendanceRecord {
        string studentId;
        string subject;
        string status;
        uint256 timestamp;
        address faculty;
    }

    mapping(string => bool) public students;
    mapping(address => bool) public authorizedFaculty;

    AttendanceRecord[] public records;

    constructor() {
        admin = msg.sender;
    }
}
```

In this code, the administrator is assigned when the contract is deployed. The students mapping tracks registered student IDs, and authorizedFaculty stores faculty wallet permissions. The AttendanceRecord structure organizes the information recorded for each attendance entry

5.2.2. Registering Students

Student registration is an important operation in the system. The administrator registers students using their student IDs. This helps maintain a list of students whose attendance can be recorded.

Sample Solidity Code:

```
function registerStudent(string memory studentId)
    external
{
    require(msg.sender == admin, "Admin only");
    require(!students[studentId], "Already registered");

    students[studentId] = true;
}
```

5.2.3. Authorizing Faculty Members

Only authorized faculty members should be allowed to mark attendance. The administrator authorizes a faculty member using their blockchain wallet address. The smart contract checks this authorization before accepting attendance transactions.

Sample Solidity Code:

```
function authorizeFaculty(address faculty)
    external
{
    require(msg.sender == admin, "Admin only");
    require(faculty != address(0), "Invalid address");

    authorizedFaculty[faculty] = true;
}
```

This function allows only the administrator to authorize faculty wallet addresses. The authorization process provides role-based access control and helps prevent unauthorized users from marking attendance.

5.3. Recording and Retrieving Attendance

5.3.1. Making Student Attendance

After faculty authorization, faculty members can record attendance for a selected subject. The faculty member provides the student ID, subject, and attendance status. The smart contract verifies the faculty's permission and checks whether the student is registered before storing the record.

Sample Solidity Code:

```
function markAttendance(
    string memory studentId,
    string memory subject,
    string memory status
) external {
    require(
        authorizedFaculty[msg.sender],
        "Faculty not authorized"
    );
    require(students[studentId], "Student not registered");
    require(
        keccak256(bytes(status)) ==
            keccak256(bytes("Present")) ||
        keccak256(bytes(status)) ==
            keccak256(bytes("Absent")),
        "Invalid status"
    );

    records.push(AttendanceRecord(
        studentId,
        subject,
        status,
        block.timestamp,
        msg.sender
    ));
}
```

5.4. Frontend and Blockchain Integration

React provides the interface for selecting attendance status and displaying records. MetaMask connects the user's wallet, and Ethers.js communicates with the smart contract.

5.5. Testing and Reports

The application is tested for student registration, faculty authorization, attendance marking, and record retrieval. The dashboard displays attendance history and calculates percentages using recorded entries for the selected subject.

The project uses a local Hardhat network for development and testing. Further testing is required before real-world deployment.

5.6. Workflow:

The workflow starts when the administrator registers students and authorizes faculty. Faculty members connect their wallets, select a subject, and mark attendance. The smart contract checks permissions and records valid attendance entries. Finally, the application displays attendance history and percentage reports.

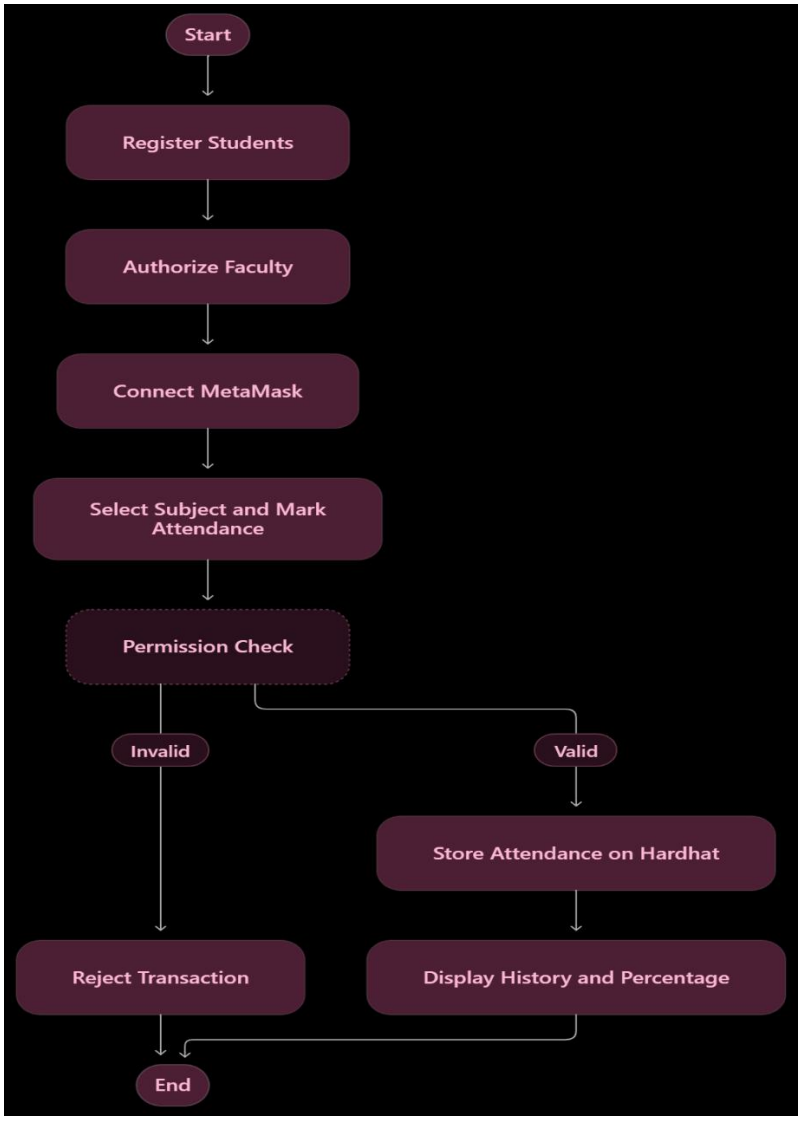


Figure 5.1: Workflow of Blockchain based Digital Warranty System

6. BENEFITS

Blockchain technology enhances student attendance management by improving data integrity, access control, transparency, and record tracking. Traditional attendance systems often rely on paper registers or centralized databases, which may lead to manual errors, unauthorized changes, and difficulties in maintaining records. By using blockchain technology and smart contracts, attendance management becomes more organized, traceable, and efficient. The key benefits of the proposed system are explained below.

6.1. Enhanced Data Integrity and Immutability

Blockchain provides a traceable record of attendance transactions. Once an attendance entry is recorded, changing the existing record is difficult under the blockchain's design. This helps reduce unauthorized modifications and improves the reliability of attendance information. The smart contract also follows predefined rules when recording attendance.

6.2. Prevention of Unauthorized Attendance

The system uses smart contract-based access control to restrict attendance operations. The administrator registers students and authorizes faculty wallet addresses. Only authorized faculty members can mark attendance, while unauthorized requests are rejected. This helps reduce the possibility of attendance being recorded by unapproved users.

6.3. Transparency and Traceability

Each attendance entry contains details such as student ID, subject, attendance status, timestamp, and faculty wallet address. These details help users review the available attendance history and trace the origin of recorded entries. This improves transparency and supports easier verification of attendance information.

6.4. Reduced Manual Errors

Traditional attendance management may involve mistakes when recording attendance or transferring details from registers to reports. The proposed system records attendance through a web application and processes it using a smart contract. This reduces some manual steps and helps maintain attendance information in an organized manner.

6.5. Easy Attendance Monitoring

The React-based frontend provides an interface for viewing student details and attendance information. Authorized faculty members can mark students as present or absent for a selected subject. The application also displays attendance history and percentage reports, making it easier to monitor student attendance.

6.6. Automated Permission Checking

The smart contract checks whether the connected wallet has the required permission before processing restricted operations. It verifies faculty authorization and student registration when attendance is submitted. These automatic checks help enforce the system's rules consistently and reduce the need for manual permission verification.

6.7. Faster Record Retrieval and Reporting

The application communicates with the smart contract through Ethers.js to retrieve available attendance information. The frontend presents attendance history and percentage reports in an organized format. This helps faculty members review records more conveniently than searching through paper registers.

6.8. Improved Accountability

The attendance record includes the faculty wallet address and timestamp associated with each entry. These details help identify which authorized wallet submitted an attendance transaction and when it was recorded. This improves accountability and supports the review of attendance history.

6.9. Cost Effective Development and Testing

The project uses Hardhat to deploy and test the smart contract on a local blockchain network. This provides a convenient development environment without requiring deployment to a public blockchain. It allows the developer to test student registration, faculty authorization, attendance marking, and record retrieval during project development.

6.10. Scope for Future Improvements

The proposed system provides a foundation for further improvements in digital attendance management. Future versions could include stronger duplicate-entry prevention, improved reporting, testing with larger student groups, and deployment to a suitable blockchain network. These enhancements could help make the system more suitable for practical educational environments.

In conclusion, the Blockchain-Based Student Attendance Management System improves attendance record organization, traceability, and permission control. By combining React, Solidity, Hardhat, MetaMask, and Ethers.js, the project demonstrates how blockchain technology can support more reliable and efficient attendance management in educational institutions.

7. CHALLENGES

Developing the Blockchain-Based Student Attendance Management System involved several challenges related to smart contracts, wallet connections, and blockchain integration.

7.1. Smart Contract Development

Writing Solidity smart contracts required careful handling of student records, faculty permissions, and attendance status.

7.2. Wallet Connection

Connecting MetaMask with the React application required proper wallet configuration and transaction approval.

7.3. Blockchain Integration

Integrating the frontend with the smart contract using Ethers.js required correct contract addresses and ABI details.

7.4. Transaction Processing

Attendance transactions needed to be processed and confirmed by the local blockchain network.

7.5. Testing and Deployment

Testing the smart contract on the Hardhat network was necessary to check attendance marking and access permissions.

7.6. User Awareness

Users need basic knowledge of wallet connections and transaction approvals to use the system correctly.

7.7 Data Accuracy

Maintaining correct student IDs, subject details, and attendance status was important to ensure that attendance records and percentage reports were accurate.

8. CONCLUSION

The Blockchain-Based Student Attendance Management System is designed to make student attendance management easier, more reliable, and more transparent. Traditional attendance systems may involve manual work, errors in record maintenance, and difficulties in tracking attendance history. This project uses blockchain technology and smart contracts to address some of these challenges and improve the process of recording and managing attendance.

The system is developed using React for the frontend, Solidity for writing smart contracts, Hardhat for the local blockchain environment, MetaMask for wallet connectivity, and Ethers.js for communication between the web application and the smart contract. These technologies work together to provide an interface through which users can interact with the attendance system.

The administrator can register students and authorize faculty wallet addresses. Authorized faculty members can mark students as present or absent for a selected subject. The smart contract checks user permissions before allowing important operations and records attendance details, including student ID, subject, attendance status, timestamp, and faculty wallet address. The web application also provides attendance history and percentage reports to help users monitor attendance records.

One of the main advantages of this project is the use of blockchain-based records, which improves traceability and helps protect recorded information from unauthorized modification through normal application operations. Smart contract-based access control also helps ensure that only authorized faculty members can mark attendance. The system reduces dependence on manual record maintenance and provides a convenient way to retrieve attendance information.

The project was implemented and tested using a local Hardhat blockchain network. During development, challenges such as smart contract creation, MetaMask configuration, frontend and blockchain integration, transaction processing, and testing were addressed. The project also provided practical knowledge of blockchain concepts, smart contracts, wallet integration, and decentralized application development.

In the future, the system can be improved by deploying it on a suitable blockchain network, adding more user-friendly features, improving report generation, and introducing additional verification mechanisms. It can also be enhanced to support larger numbers of students, multiple departments, and different academic institutions.

In conclusion, the Blockchain-Based Student Attendance Management System demonstrates how blockchain technology can be applied to academic attendance management. It provides a useful prototype for maintaining traceable attendance records, controlling access, and simplifying attendance monitoring. With further development and testing, the system has the potential to become a more comprehensive solution for educational institutions.

9. SDG's ADDRESSED

The Sustainable Development Goals (SDGs) were introduced by the United Nations to promote a better and more sustainable future. The Blockchain-Based Student Attendance Management System supports these goals by using modern technology to improve attendance management, encourage innovation, and maintain transparent records.

9.1 SDG 4: Quality Education

SDG 4 aims to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. This project supports the goal by helping educational institutions maintain organized student attendance records.

Faculty members can mark students as present or absent for selected subjects. The system also provides attendance history and percentage reports, making it easier to monitor student attendance. These features help reduce manual work and support better academic record management.

9.2 SDG 9: Industry, Innovation and Infrastructure

SDG 9 focuses on promoting innovation and developing reliable infrastructure. The project supports this goal by applying blockchain technology to attendance management.

The system uses Solidity smart contracts to record attendance and manage user permissions. React provides the user interface, while Hardhat provides a local blockchain environment for development and testing. MetaMask and Ethers.js help connect the application to the smart contract.

This project demonstrates how modern technologies can be used to develop digital solutions for educational institutions. It also provides opportunities for future improvements and further exploration of blockchain applications.

9.3 SDG 16: Peace, Justice and Strong Institutions

SDG 16 aims to promote transparency, accountability, and effective institutions. The project supports this goal by maintaining traceable attendance records and restricting important operations through access control.

The smart contract checks permissions before allowing attendance-related operations. Only authorized faculty members can mark attendance through the intended application workflow. Each attendance entry contains details such as student ID, subject, attendance status, timestamp, and faculty wallet address.

These features help users track attendance activities and improve accountability. Blockchain-based recordkeeping also helps protect recorded information from unauthorized modification through normal application operations.

10. REFERENCES

1. Solidity Documentation

Solidity is a programming language used to develop smart contracts on blockchain platforms. It was used in this project to define student registration, faculty authorization, and attendance management functions.

Available at: https://docs.soliditylang.org/

2. Hardhat Documentation

Hardhat is a development environment used to compile, deploy, and test smart contracts. In this project, Hardhat provides a local blockchain network for testing the attendance management system.

Available at: https://hardhat.org/docs

3. React Documentation

React is a JavaScript library used to build interactive user interfaces. It was used to develop the frontend through which users can interact with the attendance management system.

Available at: https://react.dev/

4. Ethers.js Documentation

Ethers.js is a JavaScript library used to connect web applications with blockchain smart contracts. It helps the React frontend communicate with the attendance smart contract and process transactions.

Available at: https://docs.ethers.org/

5. MetaMask Documentation

MetaMask is a cryptocurrency wallet that allows users to connect to blockchain applications and approve transactions. In this project, it is used to connect the user's wallet with the attendance management system.

Available at: https://docs.metamask.io/

6. Ethereum Developer Documentation

Ethereum documentation explains blockchain concepts, smart contracts, transactions, and decentralized applications. These concepts provide the technical foundation for understanding the blockchain-based attendance system.

Available at: https://ethereum.org/en/developers/docs/

7. OpenZeppelin Documentation

OpenZeppelin provides resources and tools for developing and securing smart contracts. Its documentation is useful for understanding access control and smart contract security concepts relevant to blockchain applications.

Available at: https://docs.openzeppelin.com/

8. Node.js Documentation

Node.js provides a JavaScript runtime used for development tools and JavaScript-based applications. It supports the development environment required to install and run the project's frontend and blockchain tools.

Available at: https://nodejs.org/docs/

9. MDN Web Docs – JavaScript Guide

The JavaScript Guide explains programming concepts such as functions, objects, and asynchronous operations. These concepts are useful when developing the React frontend and connecting it to the blockchain.

Available at: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide

10. United Nations – Sustainable Development Goals

The United Nations Sustainable Development Goals describe global objectives related to education, innovation, and accountable institutions. These goals provide background for the project's discussion of SDG 4, SDG 9, and SDG 16.

Available at: https://sdgs.un.org/goals

11. APPENDIX

Video presentation:

https://drive.google.com/file/d/1P7Rv95TwbOnVVwHZI8PA9cTryjhLRgVK/view?usp=sharing

Github:

https://github.com/chaitra519/BLOCKCHAIN-BASEDSTUDENTATTENDANCE-MANAGEMENTSYSTEM